Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, C/C++, SQL
Core Computer Science: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS, Multithreading
Software Development: FastAPI, Flask, REST APIs, API Integration, React.js, JavaScript, HTML, CSS, Git/GitHub
Backend & Databases: TCP/IP, Socket Programming, Redis, MySQL, SQLite, MongoDB, Database Design, Query Optimization
Data Science & Machine Learning: Machine Learning, Deep Learning, Scikit-learn, PyTorch, Pandas, NumPy, Feature Engineering, Model Evaluation, Time-Series Analysis, Anomaly Detection
Data Analytics & Visualization: Power BI, DAX, Tableau, Matplotlib, Plotly, Exploratory Data Analysis (EDA)
AI & Generative AI: Large Language Models (LLMs), RAG, Prompt Engineering, Natural Language Processing (NLP)
Tools & Engineering Practices: Linux, VS Code, Jupyter Notebook, PyTest, Unit Testing, Test-Driven Development (TDD)

EXPERIENCE

AI for Sustainability Virtual Intern **|LINK** — ***July – Sep, 2026***

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

Financial Transaction Anomaly Detection & Audit Dashboard **| LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests..

Real-Time Stock Market Anomaly Detector **|LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.

Docwise **|LINK**

- Built an AI-powered PDF question-answering system using Google Gemini and RAG, enabling contextual question answering over documents exceeding 200 pages.
- Developed a document processing and retrieval pipeline combining PDF extraction, chunking, embeddings, and semantic retrieval, achieving 95%+ query accuracy.
- Integrated FastAPI and LLM inference to deliver an end-to-end AI application for document understanding and conversational querying.

RedisLite **|LINK**

- Engineered a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) LRU cache with lazy/active TTL expiration.
- Designed a four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery.
- Validated the system with 46 automated PyTest unit tests (TDD), benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** 2026
- Deloitte Australia Data Analytics Job Simulation **|LINK** 2025